

Session 8 – Hotspot / Classification and Travel Cost / Accessibility
(Spatial Clustering Analysis and Network Analysis
with GeoDa and QGIS)

Objectives: To analysis spatial pattern of point events, to compute travel cost and measure spatial accessibility through QGIS functionalities.

Task 1: Hotspots of GP Surgeries in London Borough Haringey (Spatial Clustering Analysis)

Problem solving: Identify clusters of health facilities to support healthcare service planning

Functionality: Heatmap in QGIS

Data set: Map of GP surgeries in Haringey, Map of Haringey boundary

Add “haringey_gp.shp” and “haringey_boundary.shp”

QGIS -> Processing -> Toolbox > search “heatmap”

Choose “Heatmap (Kernel Density Estimation)”

(Please choose Radius according to the spatial extent of your study area, 500 meters is suggested in this case; Zoom to Layer of “haringey_gp”, suggested numbers of rows as 300)

Using kernel density estimation to generate a hotspot map of GP surgery in Haringey as a smooth continuous surfaces. Kernel shape as Quartic.

Crop the Heatmap with Haringey boundary

Set the colour scheme for the hotspot map

Raster -> Extraction -> Clipper Raster by Mask Layer

Layer -> Layer Properties -> Symbolology

(right click of mouse -> Properties -> Symbolology -> Render type -> Singleband pseudocolor, then change the legend)

Could you identify hotspots of GP surgeries in Haringey? What does such spatial pattern reflect?

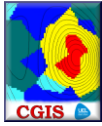
Task 2: Classifications of London Boroughs (Spatial Clustering Analysis)

Problem solving: Identify clusters of London boroughs based on health and crime characteristics to support urban planning

Functionality: Clusters in GeoDa

Data set: Deprivation, Health and Crime variable by London Borough

Input the Shape file of “london_life_polygon.shp” in GeoDa



Geoda -> Clusters -> K Means
-> Hierarchical
(Please don't tick Use geometric centroids Weighting)

Could you briefly describe classifications of London Boroughs based on deprivation, health and crime characteristics (without geometric centroids weighting)?

Task 3: The shortest path from a Tube station to a GP surgery in London Borough Haringey
(Network Analysis)

Problem solving: Compute travel cost to estimate spatial impedance to healthcare services

Functionality: Shortest Path in QGIS

Data set: Map of GP surgeries in Haringey, Map of Tube stations, Map of OS ITN road link

*Add "haringey_gp.shp", "UndergroundStations.shp" and
"Haringey_ITN_Road_Link.shp"*

QGIS -> Processing -> Toolbox > search "shortest path"
choose "Shortest path (point to point)"

(Default direction: Both directions; Speed field: speed)

-> Shortest path (Window)
Compute the shortest path from a Tube station to a GP surgery along the local road network by shortest distance (path type to calculate: Shortest) and by shortest time (path type to calculate Fastest)

Task 4: Accessibility to GP surgeries from Tube stations in London Borough Haringey
(Network Analysis)

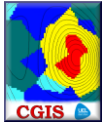
Problem solving: Measure spatial accessibility to healthcare services

Functionality: Distance Matrix in QGIS

Data set: Map of GP surgeries in Haringey, Map of Tube stations

On "haringey_gp.shp" and "UndergroundStations.shp"

QGIS -> Vector -> Analysis Tools -> Distance Matrix
(In the window of "Distance matrix",
set "Input point layer" as "Haringey_gp.shp", set "Input unique ID field" as
'UNIQUEREFE';



set "Target point layer" as *"UndergroundStations.shp"*, set "Target unique ID field" as *'STation_NA'*;
tick "Use only the nearest (k) target points" and put the number as 1,
save as "Distance.csv"

Measure relative spatial accessibility of each GP surgery to the nearest Tube station by Euclidean distance.

load "Distance.csv", join "Haringey_gp.shp" and "Distance.csv"
Properties -> Joins

creating a thematic map for accessibility of GP surgeries from underground station
-> Properties -> Symbology -> Graduated

Could you identify GP surgeries having higher accessibility to underground stations? Is there any pattern?

*Go to **SpatiaLite**, load the "Shapefile" of "Haringey_gp.shp" and "UndergroundStations.shp".
RUN following SQL commends:*

```
CREATE VIEW v1 AS
SELECT gp. Uniquerefe AS gp_surgery, gp.geometry AS gpsurgery_location,
tube.station_na AS tube_station, tube.geometry AS tubestation_location
FROM haringey_gp AS gp, UndergroundStations AS tube

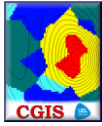
CREATE TABLE accessibility AS
SELECT gp_surgery AS "GP surgery",
       tube_station AS "Tube station",
       Min(St_Distance(gpsurgery_location, tubestation_location)) AS "Distance",
       gpsurgery_location as geometry
FROM v1
GROUP BY gp_surgery

SELECT RecoverGeometryColumn('accessibility', 'geometry', 27700, 'POINT', 'XY')
```

Come back QGIS:

QGIS -> Add SpatiaLite Layer (choose 'accessibility')

creating a thematic map for accessibility of GP surgeries from underground station
-> Properties -> Symbology -> Graduated



DS7002 Spatial Data Analysis
Practical Exercises



Please export the heatmap from **Task 1** and the accessibility map (only one map) from **Task 4** (Project -> Import/Export -> Export Map to Image), and put them into your portfolio.
Please answer questions on these two maps.